

AstroAmp RNA Kit

Cat. No. MET10010

Overview

The AstroAmp RNA Kit is a high-performance solution designed for efficient and reproducible amplification of RNA from single cells or low-input RNA samples. This kit employs an optimized reverse transcription and amplification workflow to generate full-length, high-yield cDNA suitable for downstream applications such as next-generation sequencing(NGS) and gene expression analysis.

Pack Size	48 Reactions(48 rxns)
Intended Use	For Research Use Only(RUO) * Not for diagnostic procedures
Size	92 × 92 × 57 mm
Weight	50.6 g



Key Features

- **Ultra-Sensitive RNA Amplification**
Enables efficient amplification from single cells or low-input RNA(10 pg–1 ng).
- **Full-Length cDNA Synthesis**
Captures both 5' and 3' ends for complete transcriptome coverage.
- **High-Fidelity Reverse Transcription**
Utilizes optimized RTase and TSO for enhanced efficiency and accuracy.
- **Optimized PCR Master Mix**
Delivers high reproducibility, low background and minimal bias.
- **NGS-Compatible**
Seamlessly integrates with various next-generation sequencing platforms.
- **CosmoSort-Optimized Workflow**
Designed for precise ROI sorting with minimized lysis buffer volume to enhance reaction efficiency and preserve sample integrity for high-quality data acquisition.

Applications

- Single-cell RNA sequencing(scRNA-seq)
- Low-input RNA analysis
- Gene expression profiling
- cDNA library preparation for NGS
- A-to-I edited mutants detection
- mRNA isotypes identification

Ordering Information

- **Catalogue Number:** MET10010
- **Package Size:** 48 reactions
- **Manufacturer:** Meteor Biotech Co.Ltd.

Storage & Stability

- Store all reagents at **-20°C**.
- Avoid repeated freeze-thaw cycles.
- Ensure components are protected from RNase contamination.

Performance Advantage

Amplification Plot

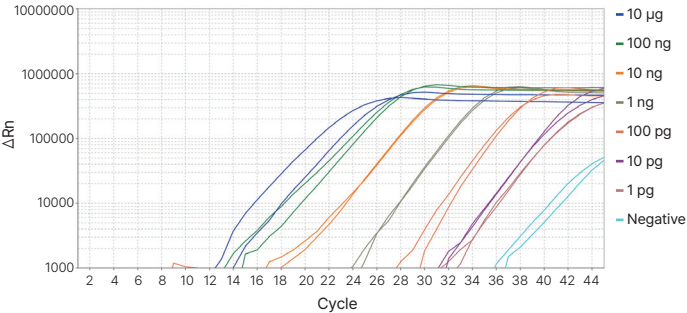


Figure 1. AstroAmp RNA kit enables efficient cDNA synthesis from RNA inputs as low as 1 pg using TBP primers. The RTase shows high sensitivity and robust performance across a broad input range. No signal is detected in the negative control.

Amplification Plot

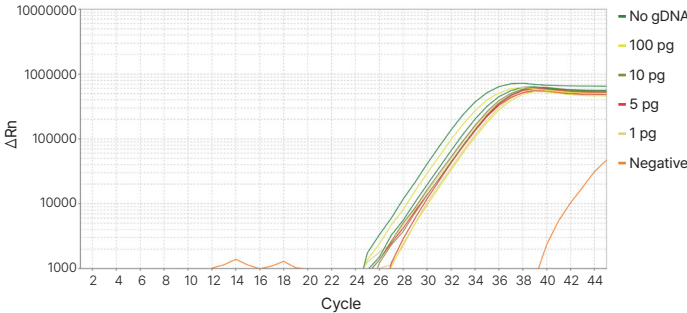


Figure 2. AstroAmp RNA kit enables efficient cDNA synthesis across RNA inputs (100 pg to 1 pg) using TBP primers. Even with gDNA contamination, the RTase performs reliably without interference. No amplification is observed in the negative control, confirming specificity.

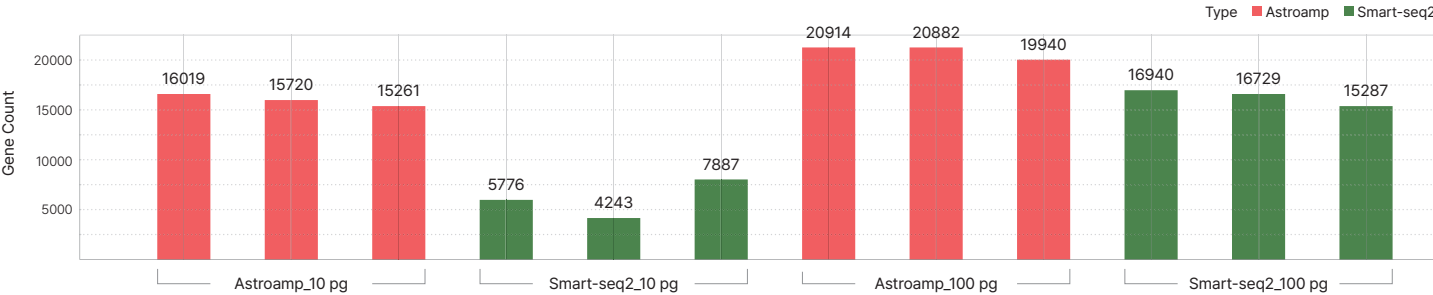


Figure 3. The AstroAmp RNA kit consistently outperforms Smart-seq2 in gene detection across picogram-scale RNA inputs. At 10 pg, AstroAmp captures substantially more genes, highlighting its high sensitivity and reliability for transcriptome profiling of scarce or degraded samples.

Gene Coverage Analysis

Astroamp vs Smart-seq2

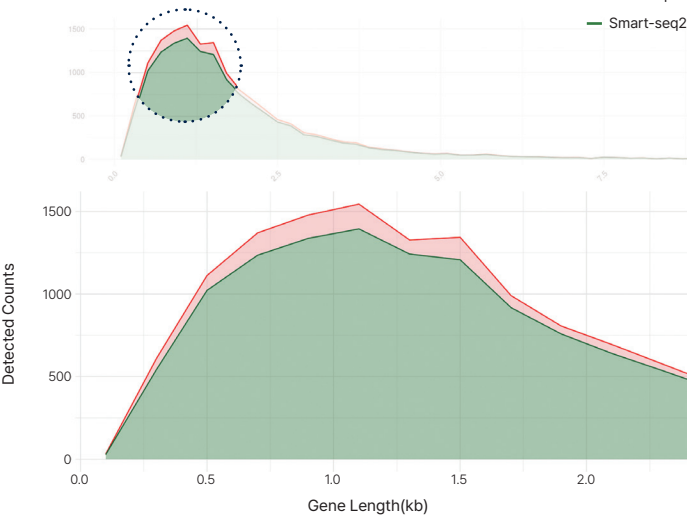


Figure 4. Gene coverage analysis. This figure evaluates how thoroughly genes of different lengths are represented in a sequencing experiment. “Coverage” refers to the number of genes detected or the proportion of a gene’s sequence that is captured by sequencing reads.

Correlation Matrix Analysis

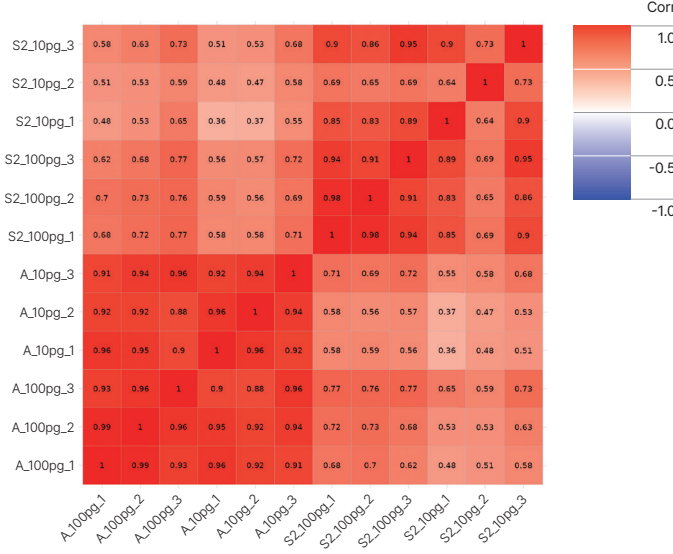


Table 1. Correlation Matrix Analysis. This analysis is a table displaying the pairwise correlation coefficients between different variables. In this context, each entry indicates the strength and direction of the linear relationship between two datasets (e.g., A_10pg_1, S2_100pg_3). The coefficients typically range from -1(perfect negative correlation) to 1(perfect positive correlation), with values closer to zero indicating weaker relationships.